

7(a)	${}^{66}_{29}\text{Cu}$	B1
7(b)(i)	the energy of the decay is fixed / constant	B1
	the energies of the beta particles have a (continuous) range of values / varies / not constant	B1
	another particle / an (anti)neutrino must possess the extra / remaining energy (difference between energy of the decay and the β kinetic energy)	B1
7(b)(ii)	(electron) antineutrino	B1
7(b)(iii)	${}^{66}_{29}\text{Cu} \rightarrow {}^{66}_{30}\text{Zn} + {}^0_{-1}\beta + {}^0_0\overline{\nu}_e$ values for Cu and Zn correct with no other extra particles on either side of the equation	B1
	second term correct (${}^0_{-1}\beta$)	B1
	third term correct (${}^0_0\overline{\nu}_e$)	B1